Web Engineering & Development (SWE 363)

Introduction to Web Development

Announcements 📣

- Clone the course repository from GitHub
- Join Blackboard Common section
- Start looking for a project team (3-4 students)

Today's plan:

We will get to know the web, explore the journey of an HTTP request, and build a simple web application.

- The world wide web (WWW)
- Infrastructure of the web
- HTTP Requests and Responses
- IP addresses
- DNS (Domain names system)
- URLs (Uniform Resource Locator)
- intro to HTML, CSS & JS

So ..

What is the web?

aka World Wide Web or WWW

The web

*A vast network of interconnected documents and resources, accessible via the **internet**, that allows users to share and access information globally.*



What technologies makes the web possible?

What technologies makes the web possible?

Internet

HTTP

Servers

Clients

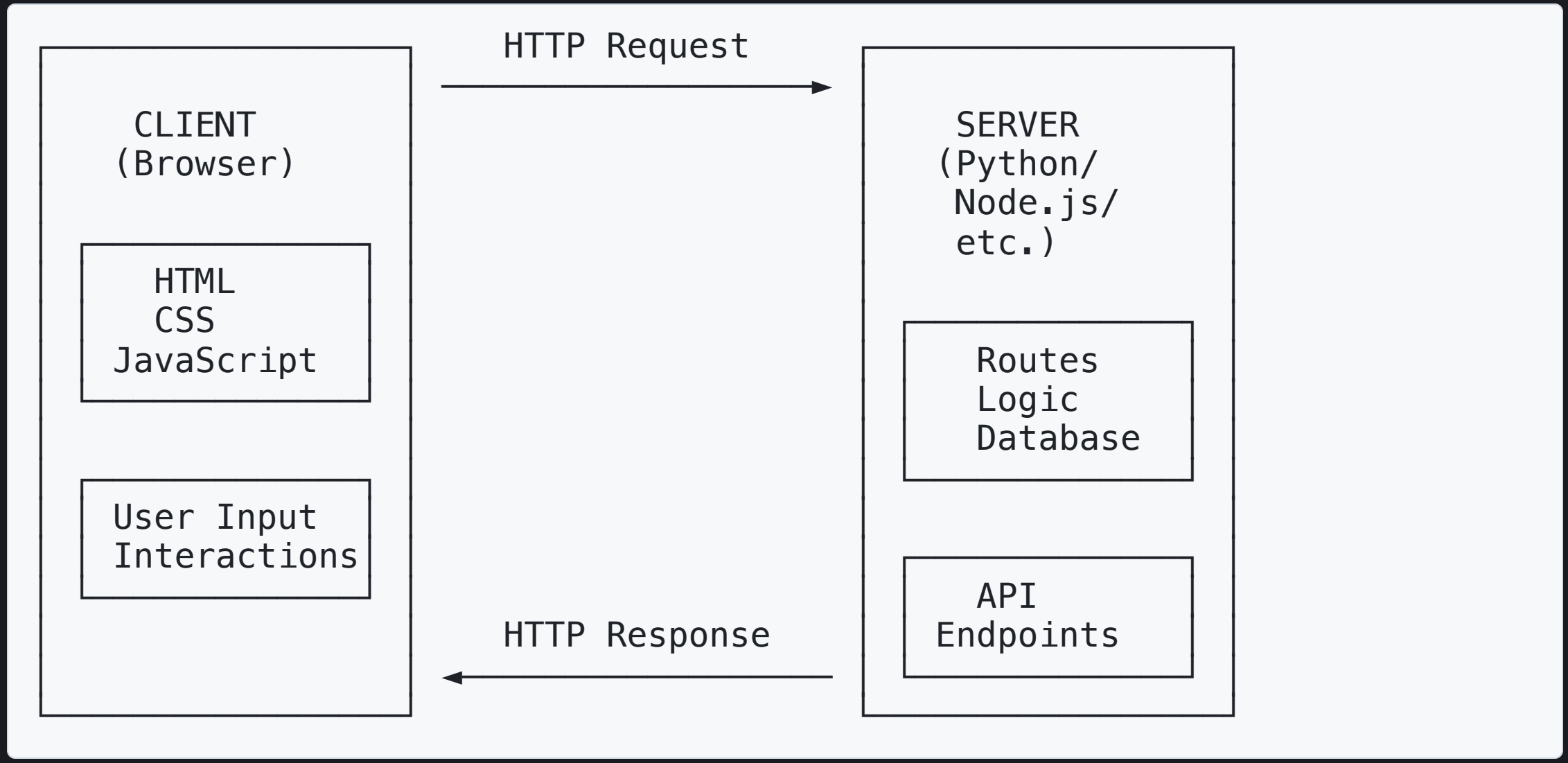
HTML

URLs

What makes a web application?

A web application is a software application that runs on a **web server** and is accessed through a **web client**.

Architecture of a simple web app



~10m

Explore it yourself

- Go to google.com
- Right click > inspect
- Search for any keyword (e.g. KFUPM)
- Notice: # requets, requests latecny, type of resource requested, size of response)

What happens when we Google "KFUPM"?

- Where does our request be **directed**?
- How does it know the **path**?
- How does it ask for specific **results**?
- How are the results **displayed**?

The Journey of a Web Request 🌐

1. User types **URL**
2. **DNS** Lookup
3. **IP Address** Found
4. HTTP **Request** Sent
5. HTTP **Response**
6. Browser **Renders** Page

1. User types a URL

https://www.google.com/search?q=kfupm

- Shceme
- Hostname
- Path
- Query string

2. DNS Lookup

- **Browser** checks its **cache** first
- If not found, check **OS cache**
- If not found, check **router cache**
- If not found, ask **ISP DNS server**
- **ISP** asks **Root DNS servers**
- **Root** points to **TLD servers** (.com)
- **TLD** points to **authoritative server**
- **Authoritative server** returns **IP address**
- **Result** is **cached** at all levels

3. IP Address found

Google.com -> 172.217.18.36

- A unique number that **identifies** each computer using the Internet Protocol to communicate over the internet

4. HTTP Request Sent

HTTP REQUEST
GET /search?q=KFUPM HTTP/1.1 Host: google.com User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) Accept: text/html,application/xhtml+xml,application/xml Accept-Language: en-US,en;q=0.9 [Empty body for GET request]

5. HTTP Response Recieved

HTTP RESPONSE

```
HTTP/1.1 200 OK
Date: Mon, 26 Aug 2024 12:34:56 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 10240
Connection: keep-alive

<!DOCTYPE html>
<html>
  <head><title>Search results for KFUPM</title></head>
  <body>... (HTML content) ...</body>
</html>
```

6. Browser Renders Page

Response is returned in HTML, CSS and JS

A Simple HTML Page

```
<!doctype html>
<html>
  <head>
    <style>
      body { background-color: lightblue; }
    </style>
  </head>
  <body>
    Hello, world!
    <script>
      console.log('Hello from JavaScript!');
    </script>
  </body>
</html>
```

Demo ~15m

Build a simple HTML page

- Go to our repo (repo link is also found in BB)
- Go to exercises guidelines folder and open 1.1 Hello HTML
- Read the instrutions in the README.md file

Demo ~15m

Building a real web app

- Go to our repo (repo link is also found in BB)
- Go to exercises guidelines folder and open 1.2 Hello Web
- Fork the repository to your own Github account
- Read the instructions in the README.md file
- Complete the code and push to Github

Next class:

- HTML Document
- Basic HTML Tags
- HTML Lists
- HTML Tables
- HTML Images